Decision-Centric Probabilistic Forecasting for Inventory Optimization

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Outline

1. Why Probabilistic Forecasts?
2. Techniques for Probablistic Forecasting
3. Demand during Lead Time + Review Period
4. Stochastic Inventory Optimization
   - Hierarchical Newsvendor for Substitutable Demand
5. Conclusion

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Point Forecasts

Point forecasts predict expected demand: $\hat{D}_{i,\tau} \approx \mathbb{E}[D_{i,\tau}|x_{i,t}]$, where:

- i = SKU / product
- τ = future time period being forecasted
- t = decision time / forecast origin
- $D_{i,\tau}$ = random demand for SKU i at time τ
- $x_{i,t}$ = feature vector available at time t for forecasting demand in period τ. Example:

$$x_{i,t} = \begin{bmatrix} \text{day of week}_\tau \\ \text{holiday flag}_\tau \\ \text{price}_{i,\tau} \\ \text{promotion}_{i,\tau} \\ \text{demand lags up to } t \\ \text{rolling demand features up to } t \\ \text{SKU / category / store attributes} \\ \vdots \end{bmatrix}$$

But inventory decisions depend on **risk**, not only expected demand.

The Newsvendor Problem I

- The Newsvendor Problem is a mathematical model used to determine the optimal inventory level for perishable or single-period goods.
- It balances the cost of overstocking (unsold items) against the cost of understocking (lost profit from unmet demand) to minimize expected total cost.
- The cost of over-forecasting and under-forecasting is asymmetric:

$$C(q, D) = h(q - D)^+ + b(D - q)^+$$

where h is overage / holding / spoilage cost and b is underage / lost-sales / service cost.

- The optimal inventory is a quantile, not the mean:

$$q^* = F^{-1}\left(\frac{b}{b+h}\right)$$

where F is cumulative distribution function (cdf) of the demand.

The Newsvendor Problem II

- For high service levels (95%-99%), the right tail of the demand distribution is crucial for determining the optimal inventory level.
- Two SKUs can have the same mean but very different safety stock requirements:

$$\mathbb{E}[D_1] = \mathbb{E}[D_2], \quad F_1^{-1}(0.95) \neq F_2^{-1}(0.95)$$

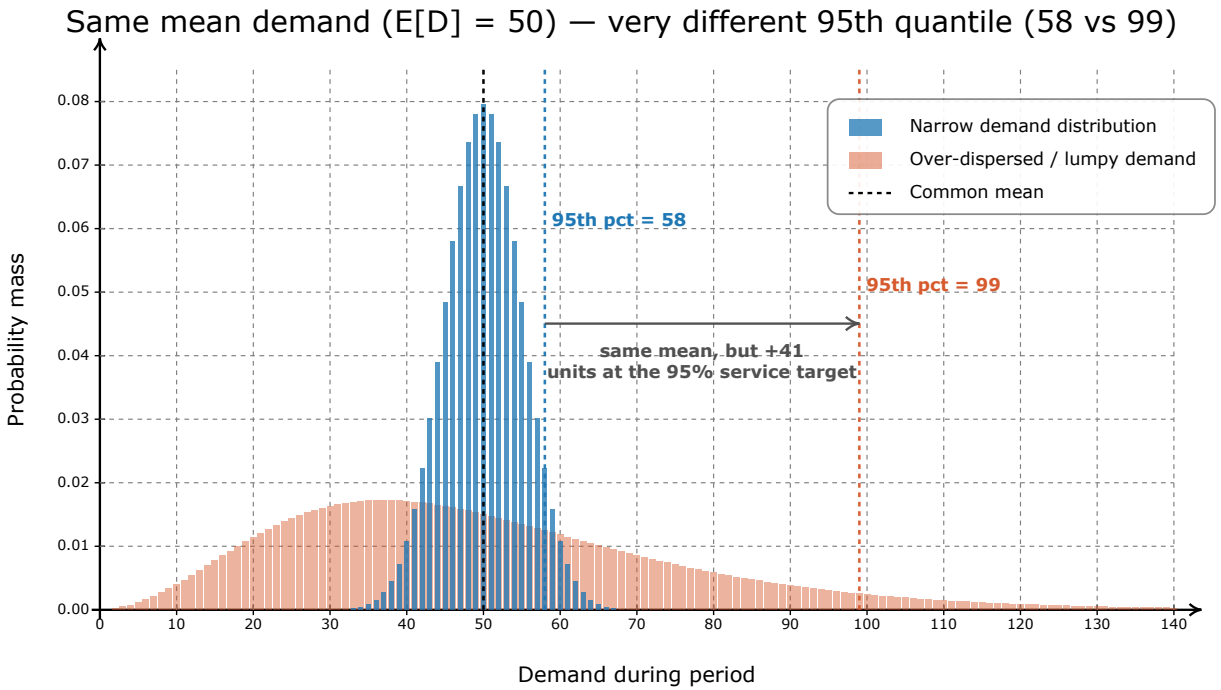


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Generalized Linear Models

A Generalized Linear Model (GLM)[1] is a generalization of linear regression that models the conditional distribution of a response variable, Y, given a vector of features, X.

- **Exponential Family:** The conditional distribution of the response variable Y, given X, usually belongs to the exponential family, for example Poisson or Negative Binomial. Its conditional mean is denoted as $\mathbb{E}[Y|X] = \mu$.
- **Link Function:** A monotonic, differentiable function $g(\cdot)$ that bridges the linear predictor to the expected value of the distribution: $g(\mu) = X\beta$.
- **Output:** A customized probability distribution tailored to that specific input's features, X.
- GLMs don't use ordinary least squares (OLS) like linear regression. Instead, they rely on maximum likelihood estimation (MLE).

[1] Original formulation by Nelder and Wedderburn 1972.

Poisson Regression I

- **Probability Mass Function:** Models the probability of observing y units of demand given a mean rate μ: $\mathbb{P}(Y = y \mid X) = \frac{e^{-\mu}\mu^{y}}{y!}$
- **Canonical Link:** Uses the log link function $\ln(\mu) = X\beta$, guaranteeing strictly positive demand predictions $\mu = e^{X\beta}$.
- **Equidistribution Property:** For Poisson distribution, the mean equals the variance: $\mathbb{E}(Y) = \mathrm{Var}(Y) = \mu$
- **Likelihood Function:** The joint probability of n independent demand observations:

$$L(\beta) = \prod_{i=1}^{n} \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}$$

Poisson Regression II

- **Log-Likelihood Objective:** Taking the natural log yields the function maximized during optimization:

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i (X_i \beta) - e^{X_i \beta} - \ln(y_i!) \right]$$

- **Parameter Estimation:** Solved using Iteratively Reweighted Least Squares (IRLS) since no closed-form solution exists.

Negative Binomial Regression I

- **Poisson Failure:** Real-world inventory data exhibits overdispersion ($\text{Var}(Y) > \mathbb{E}(Y)$) due to bulk buying, promotion spikes, or unobserved stockouts.
- **Underestimated Risks:** Applying Poisson to overdispersed data underestimates standard errors, leading to dangerously low safety stock levels.
- **Probability Mass Function:** The Negative Binomial distribution is parameterized by:

$$\mathbb{P}(Y = y \mid \mu, \alpha) = \frac{\Gamma(y + \alpha^{-1})}{y!\,\Gamma(\alpha^{-1})} \left(\frac{\alpha^{-1}}{\alpha^{-1} + \mu}\right)^{\alpha^{-1}} \left(\frac{\mu}{\alpha^{-1} + \mu}\right)^{y}$$

- **Flexible Variance:** The relationship between mean and variance can be expressed using the dispersion parameter α: $\text{Var}(Y) = \mu + \alpha\mu^2$

Negative Binomial Regression II

- **Log-Likelihood Objective:** The optimization function used to simultaneously estimate regression coefficients β and dispersion α:

$$\ell(\beta, \alpha) = \sum_{i=1}^{n} \Big[\ln \Gamma(y_i + \alpha^{-1}) - \ln \Gamma(\alpha^{-1}) - \ln(y_i!) - \alpha^{-1} \ln(1 + \alpha e^{X_i \beta}) + y_i \ln(\alpha e^{X_i \beta}) - y_i \ln(1 + \alpha e^{X_i \beta}) \Big]$$

- **Parameter Estimation:** Solved using L-BFGS.
- In Python, the easiest way to fit GLMs is using the `statsmodels` package.

Generalized Additive Models for Location, Scale, and Shape (GAMLSS)

GAMLSS[2] is a parametric framework that models every parameter of a distribution - not just the mean.

- GAMLSS allows independent modeling of up to four distribution parameters: Location (μ), Scale (σ), Shape 1 (ν, skewness), and Shape 2 (τ, kurtosis).
- Each distribution parameter can be linked to its own unique set of predictors and covariates via specific link functions.
- The framework is not locked into the exponential family.
- Ulrich et al. (2021) used GAMLSS to arrive at the cost-minimising solution to the newsvendor model in an e-grocery setting.

[2] Introduced by Rigby and Stasinopoulos 2005.

Quantile Regression I

- The **pinball loss function**[3] measures the accuracy of models that predict specific percentiles of the distribution rather than the average.

$$L_q(y, \hat{y}) = \begin{cases} q(y - \hat{y}) & \text{if } y \geq \hat{y} \quad \text{(Underestimation)} \\ (1 - q)(\hat{y} - y) & \text{if } y < \hat{y} \quad \text{(Overestimation)} \end{cases}$$

- **Model Agnostic Architecture:** Quantile regression is not limited to linear models. The objective function can be optimized using CatBoost, XGBoost, Deep Neural Networks, or any gradient-based learner.
- **Multi-Quantile Regression:** To get an accurate picture of the conditional distribution, we train models to simultaneously predict a full set of quantiles (e.g., 0.1, 0.2, ..., 0.9, 0.95, 0.99).
  - We don't need to train a separate model for each quantile.
  - Instead of producing one number, the model produces a vector of K numbers, where K is the number of target quantiles.

Quantile Regression II

- This objective function is the summed pinball loss evaluated across all specified quantiles simultaneously:

$$L_{\text{MultiQuantile}} = \sum_{k=1}^{K} L_{q_k}(y, \hat{y}_{q_k})$$

- The Cumulative Distribution Function (CDF) and the Probability Density Function (PDF) can be reconstructed from the grid of quantiles without parametric assumptions.

[3] First proposed by Koenker and Bassett 1978.

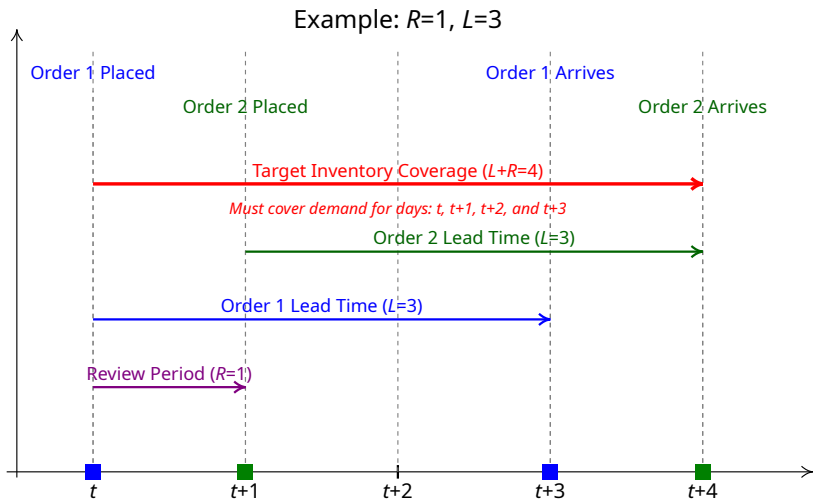
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Periodic Review Inventory Policy with Lead Time

Many real-world inventory management systems follow a periodic review policy:

- **Reorder interval** (R): The number of time periods between two consecutive orders.
- **Lead time** (L): The number of time periods an order takes to arrive.
- To avoid stockouts, the target inventory at period t must cover the cumulative demand over the $L + R$ exposure window[4]: $D_t, D_{t+1}, \ldots, D_{t+L+R-1}$.



[4] See for example "Stochastic Inventory Models: Periodic Review" 2019.

Distribution of Demand during Exposure Window $(L + R)$

The Newsvendor model can be extended to accommodate positive lead times and periodic reviews.

- **Key Idea**: Optimize for the total demand during the exposure period $(L + R)$, rather than a single period:

$$D_t^{L+R} = D_t + D_{t+1} + \cdots + D_{t+L+R-1}$$

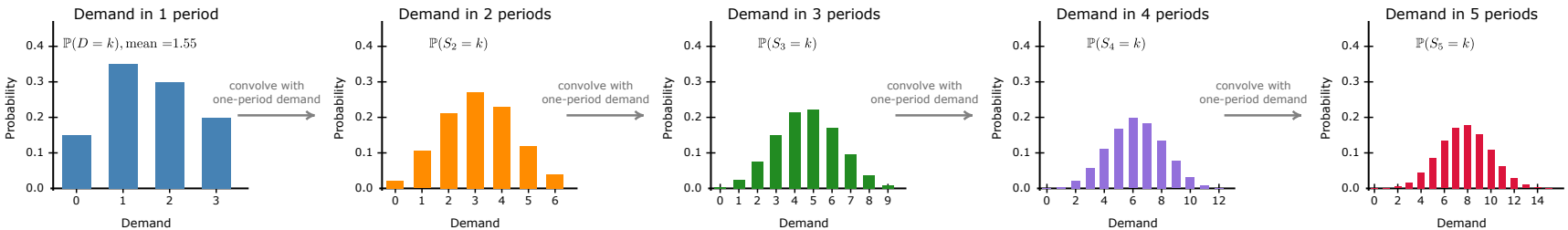
- Assuming independent periods, the probability distribution of D_t^{L+R} is computed by convolving the individual demand distributions.
- If the lead time (L) is stochastic, the distribution of D_t^{L+R} becomes a mixture distribution. It weights the cumulative demand $(S_{t,l+R} = D_t + \cdots + D_{t+l+R-1})$ across all possible lead times up to a maximum $\bar{L}$:

$$\mathbb{P}(D_t^{L+R} = k) = \sum_{l=1}^{\bar{L}} \mathbb{P}(S_{t,l+R} = k)\mathbb{P}(L = l)$$

Example: Convolution-based computation of Demand during Lead Time + Review Period

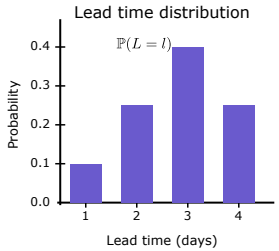
Precompute horizon-demand distributions, then mix them using the lead-time distribution.

Step 1: Repeated convolution of the one-period demand distribution
Compute and store $\mathbb{P}(S_h = k)$ up to h = Max Lead Time ($\bar{L}$) + Review Period (R)



Step 2: Compute the distribution of demand during lead time + review period (D^{L+R}).

precomputed horizon distributions



lead-time weights

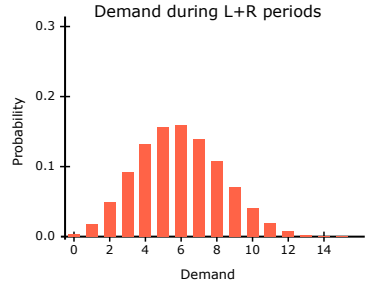
For each realized lead time l, we have
$\mathbb{P}(D^{L+R} = k \mid L = l) = \mathbb{P}(S_{l+R} = k)$

Therefore, by conditional probability:

$$\mathbb{P}(D^{L+R} = k) = \sum_{l=1}^{\bar{L}} \mathbb{P}(S_{l+R} = k)\mathbb{P}(L = l)$$

Here: $R = 1$, L takes values 1, 2, 3, 4, so horizons 2, 3, 4, 5 are mixed.

mixture / compound distribution



Example distributions used only for illustration: one-period demand support {0,1,2,3}; lead time support {1,2,3,4}.

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Multi-Product Newsvendor with Budget Constraint

- A decision-maker needs to determine the target inventory for N distinct products while satisfying a working capital constraint[5].
- The total capital tied up in the target inventory cannot exceed a predefined working capital limit, denoted as B.
- Inputs:
  - Demand during lead time + review period (D_i): Random variable with CDF $F_i(\cdot)$
  - Economic parameters for product i:
    - Selling price of product i: p_i per unit
    - Purchase cost of product i: c_i per unit
    - Shortage penalty of product i: r_i per unit.
  - Available budget: B

[5] Zhang, Xu, and Hua (2009) considered a similar formulation. Perakis, Singhvi, and Spantidakis (2020) further extended this to multi-item, multi-warehouse settings under dual capacity constraints.

Problem Formulation

- The profit from the sales of product i, denoted as $P_i(x_i)$, balances revenue against purchase costs and shortage penalties for the target inventory x_i:

$$P_i(x_i) = p_i \min\{D_i, x_i\} - c_i x_i - r_i \max\{D_i - x_i, 0\}$$

- Goal: Maximize the total expected profit across all N products:

$$P(x_1, \cdots, x_N) = \sum_{i=1}^{N} \mathbb{E}[P_i(x_i, D_i)]$$

- The Constraint: The total working capital required to fund the target inventory must not exceed the available limit B:

$$\sum_{i=1}^{N} c_i x_i \leq B$$

Solving via Binary Search I

- We relax the hard working capital constraint by introducing a non-negative Lagrange multiplier, $\lambda \geq 0$. This translates the constrained problem into an unconstrained maximization:

$$\text{Maximize } \mathcal{L}(x_1, \cdots, x_N, \lambda) = \sum_{i=1}^{N} \mathbb{E}[P_i(x_i, D_i)] - \lambda \left(\sum_{i=1}^{N} c_i x_i - B \right)$$

- This is equivalent to a standard newsvendor problem where the effective cost is adjusted to $c_i' = c_i + \lambda$. The optimal $x_i^*(\lambda)$ is given by:

$$x_i^*(\lambda) = F_i^{-1} \left(\frac{p_i - c_i(1+\lambda) + r_i}{p_i + r_i} \right)$$

- Consequently, the total working capital tied up by this optimal solution, $C(\lambda) = \sum_i^N c_i x_i^*(\lambda)$, acts as a non-increasing function of λ.

Solving via Binary Search II

- Our objective is to find the smallest value of $\lambda \geq 0$ such that the total cost $C(\lambda)$ satisfies the working capital constraint $C(\lambda) \leq B$.
- Because $C(\lambda)$ is monotonic with respect to λ, we can efficiently locate the optimal λ using a binary search.
- The Procedure:
  - First, check the unconstrained solution ($\lambda = 0$). If $C(0) \leq B$, then $x_i^*(0)$ is the optimal order quantity for all products.
  - If the budget is exceeded, initiate a binary search over the interval $[0, \lambda_{\max}]$, where $\lambda_{\max} = \min_{i=1,\cdots,N} \left(\frac{p_i + r_i}{c_i} - 1 \right)$.
  - Set $\lambda_L = 0$, $\lambda_U = \lambda_{\max}$, and evaluate the midpoint $\lambda_M = (\lambda_U + \lambda_L)/2$.
  - If $C(\lambda_M) > B$, then λ_M is too small; continue the search in $[\lambda_M, \lambda_U]$.
  - If $C(\lambda_M) \leq B$, then λ_M is potentially feasible but might be too large; continue the search in $[\lambda_L, \lambda_M]$.

Demand Substitution & Risk Pooling

- In practice, products within a parent group or Master SKU (MSKU)—such as sizes, colors, or close variants—often exhibit **demand substitution** when a preferred SKU stocks out.
- Exact multi-product newsvendor models under substitution[6] require solving coupled non-linear optimality conditions with no closed-form solution.
- **Key Insight:** Substitution acts as a **risk-pooling** mechanism across child SKUs within an MSKU M:
  - Optimizing each SKU $i \in M$ independently ignores substitution and over-allocates safety stock across every variant.
  - Since aggregate MSKU variance is smaller than the sum of SKU-level variances, pooling demand at the MSKU level reduces total buffer inventory while maintaining the same group service level.

[6] Mahajan and van Ryzin (2001) and Netessine and Rudi (2003) characterize optimal stocking under dynamic and static substitution, while Rajaram and Tang (2001) and Kök and Fisher (2007) develop tractable approximations for retail assortments.

Two-Stage Hierarchical Decomposition I

We decompose the optimization across G MSKUs $(M_1, \ldots, M_G)$ into two tractable multi-product newsvendor stages, both solved via binary search:

- **Stage 1: MSKU-Level Optimization (Budget Allocation)**
  - Let $D_M = \sum_{i \in M} D_i$ be the aggregate MSKU demand during the exposure window with CDF $F_M(\cdot)$ (computed via convolution or direct MSKU forecasting).
  - Define MSKU unit economics using demand-share weights $w_i = \frac{\mathbb{E}[D_i]}{\sum_{j \in M} \mathbb{E}[D_j]}$:

$$\bar{p}_M = \sum_{i \in M} w_i p_i, \quad \bar{c}_M = \sum_{i \in M} w_i c_i, \quad \bar{r}_M = \sum_{i \in M} w_i r_i$$

  - Solve the budget-constrained newsvendor problem across all MSKUs subject to $\sum_M \bar{c}_M Q_M \leq B$ to obtain the pooled target inventory Q_M^*:

$$Q_M^*(\lambda) = F_M^{-1}\left(\frac{\bar{p}_M - \bar{c}_M(1+\lambda) + \bar{r}_M}{\bar{p}_M + \bar{r}_M}\right)$$

Two-Stage Hierarchical Decomposition II

- **Stage 2: Within-MSKU SKU Allocation (Quantity Cap)**
  - For each MSKU M, disaggregate the pooled target Q_M^* across its child SKUs $i \in M$ by solving a sub-newsvendor problem under a **total unit constraint**:

$$\max_{\{x_i\}_{i \in M}} \sum_{i \in M} \mathbb{E}[P_i(x_i, D_i)] \quad \text{subject to} \quad \sum_{i \in M} x_i \leq Q_M^*$$

  - Introducing a group-specific dual multiplier $\mu_M \geq 0$ on unit capacity yields the closed-form SKU quantile rule:

$$x_i^*(\mu_M) = F_i^{-1}\left(\frac{p_i - c_i - \mu_M + r_i}{p_i + r_i}\right)$$

  where μ_M is found via binary search such that $\sum_{i \in M} x_i^*(\mu_M) \leq Q_M^*$.

Two-Stage Hierarchical Decomposition III

- **Partial Substitution ($\theta \in [0,1]$):** If only a fraction θ of unmet demand substitutes within M, interpolate the cap between the independent sum $Q_M^{(0)} = \sum_{i \in M} x_i^*(\lambda)$ and the fully pooled target $Q_M^{(1)} = Q_M^*(\lambda)$:

$$Q_M(\theta) = (1-\theta) Q_M^{(0)} + \theta Q_M^{(1)}$$

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Takeaways

- Forecasts are not the objective, inventory decisions are.
- Probabilistic forecasting estimates the entire conditional demand distribution: $\mathbb{P}(D_{t+\tau} \mid x_t)$
- Lead time and review period uncertainty are propagated by computing the distribution of cumulative demand:

$$D_t^{L+R} = D_{t+1} + \cdots + D_{t+L+R}$$

- Inventory targets are obtained by solving a stochastic optimization problem that explicitly accounts for
  - service level requirements,
  - holding and shortage costs,
  - and working capital constraints.

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